

Research papers

Adaptive droop-based SoC balancing control scheme for parallel battery storage system in shipboard DC microgrid

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ABSTRACT

In this article, an adaptive droop control strategy is proposed for parallel battery storage systems (BSSs) in shipboard DC microgrids, addressing critical challenges such as State-of-Charge (SoC) equilibrium, precise load power distribution, and regulation of DC bus voltage. In the primary control layer, an innovative adaptive droop-based SoC (ADBS) controller is introduced, leveraging SoC information with dynamic droop coefficient adaptation based on a sigmoid function. ADBS aims to stabilize SoC dynamically while ensuring accurate load current allocation among battery storage units (BSUs). A virtual voltage drop equalization controller (VVDE) is employed to mitigate the influence of line impedance mismatches, while a bus voltage compensation controller (BVC) is utilized to compensate for DC bus voltage deviation in the secondary control layer. Communication is established through a sparse network in the communication layer, where each BSU communicates exclusively with its adjacent BSUs. An iterative multi-agent consensus algorithm collectively estimates the average values of global variables for communication burden reduction and coordinated control efforts. Additionally, stability analysis for the suggested control approach is conducted. Finally, a simulation model (MATLAB/Simulink) and experimental platform (Star Sim HIL) have been developed to validate the strategy's effectiveness. The results of the proposed approach successfully attain dynamic SoC equilibrium, precise load current distribution, and DC bus voltage restoration in diverse environments, with notably faster SoC convergence compared to state-of-the-art methods. More importantly, through comparative analysis, it is evident that the presented approach demonstrates significantly superior performance. The SoC and current of each BSU have ultimately attained consistency, with errors approximately around 0 % and 0 A. In contrast, the comparison method yields errors of approximately 3 % and 1.02 A in the SoC and current of each BSU in the simulation results, and around 4 % and 2 A in the case of experimental results. Finally, the proposed methodology offers a comprehensive solution, showcasing robust performance, adaptability, and superior control in diverse operating conditions.

1. Introduction

With the depletion of worldwide fossil fuels and the growing concerns about environmental sustainability, emission-reduction, and energy-saving technologies have gained significant attention in the shipping industry [1–3]. Over the past ten years, the shipping industry has calculated for approximately 90 % of global transportation logistics for international trade [4], and it is responsible for around 3 % to 5 % of the overall worldwide greenhouse gas (GHG) emissions [5]. Moreover, research conducted by the International Maritime Organization (IMO) has indicated that if immediate action is not taken, this percentage could skyrocket to 18 % by 2050 due to the rapid increase in overseas trading [6]. Due to fossil energy crisis and greenhouse effect, microgrids using

renewable energy sources (RESs) have rapidly emerged and are considered to provide a solution to the issues [7–9]. As reactive power, harmonics, phase synchronization, and other challenges are not pertinent to a DC microgrid (DC-MG), it has garnered greater attention in recent year in comparison to an AC microgrid [10]. Furthermore, A DC microgrid operates highly efficiently and is versatile due to the minimal power conversion steps required by most RESs and BSS when linked to a DC bus [11–13]. As a result, there is growing interest in implementing DC microgrids in modern ship energy systems for energy utilization.

However, unlike land-based DC microgrids that can function both mobile mode and connected to the AC grid, DC shipboard microgrids (DC-SM) solely operate off-grid and cannot transfer energy with the AC grid. Additionally, new energy generation systems on ships, like

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photovoltaic (PV) and wind energy schemes are impacted not only due to natural elements but also because of the intricate navigation circumstances of vessels. Consequently, their power production fluctuates often, presenting significant challenges to system stability. For the smooth operation of the DC-SM, incorporating BSUs is essential. These BSUs play a crucial role in mitigating power fluctuations from new energy sources, maintaining energy equilibrium, and ensuring overall system stability [14].

To enhance power quality and stability of system, BSS often connected with the DC microgrid in parallel, and typically utilized for peak shaving and valley filling applications [15]. Nevertheless, differences in the SoC values among BSUs and inaccuracies in power sharing due to line impedance can result in SoC imbalances. These imbalances have the potential to shorten the operational lifespan of the BSUs [16,17]. Furthermore, these discrepancies could potentially impact the reliable execution of the DC-SM, especially because of the early disconnection of specific BSUs. Consequently, it becomes imperative to explore a dynamic equalization control approach for BSUs on board ships, while taking their individual SoCs into account. This research aims to achieve both autonomous and rational distribution of output current among each paralleled converter and facilitate rapid and stable SoC equilibrium for every BSUs [18]. In addition, voltage of DC bus is an essential criterion to assess system stability of the DC-MG, ensuring its stability close to the rated value in real time during ship navigation. As a result, the secondary control challenge in DC-MG is the regulation of the DC bus voltage [19]. Various techniques for controlling DC bus voltage have been suggested and are commonly categorized into two categories: droop control strategies and master-slave control strategies [20]. The master-slave control system's reliance on a single master converter is a notable drawback. Any malfunction in this converter would lead to a complete system shutdown. In contrast, droop control strategies offer reliability by enabling cooperative operation between paralleled converters without a communication link, eliminating the single point of failure issue [21]. Droop control is more dependable and cost-efficient than the master-slave approach. Consequently, droop control schemes are gaining prominence in DC-MG and have been widely studied [22].

To address the challenge of SoC imbalance, researchers have introduced various control strategies. The approaches to achieving SoC balance fall under three categories: decentralized [23–25], centralized [26], and distributed [14,27–31], depending upon their application of communication approach. A decentralized approach for achieving dynamic SoC balancing with parallel BSSs is introduced in [23], utilizing a droop-based methodology. This method attains both SoC equilibrium and power sharing through the proportional adjustment of the droop coefficient based on the SoC's n-order. However, this approach may lead to a substantial drop in the bus voltage. Ref. [24] introduces decentralized control techniques. Through the adjustment of the droop coefficient, BSUs with lower SoC levels deliver reduced power, while units with higher SoC levels provide increased power, effectively achieving automatic SoC balance. Ref. [25] introduces a decentralized approach employing fuzzy logic to guarantee both SoC balance and DC bus voltage recovery across BSUs. Despite their independence from communication, these techniques exhibit limited SoC balancing accuracy due to the absence of information exchange among BSUs. To address stability concerns arising from droop coefficient adjustments, a novel strategy is presented in [26]. This approach achieves SoC equilibrium through voltage shifting, using a coupling factor “X” that encompasses global SoC information. SoC values converge as “X” is adaptively adjusted to shift the droop curve. However, the centralized control method, while offering higher accuracy, relies heavily on the communication network and a centralized controller to achieve its objectives. In the face of a network attack or a single point of failure, the entire communication network may be compromised.

Researchers have suggested distributed control-based approaches to address the problems associated with decentralized and centralized approaches regarding both accuracy of SoC balance and

communication, respectively. In Ref. [27], a PI controller calculates the droop coefficient to guide each BEU's SoC value towards an average. The inclusion of virtual power in the voltage shifting term ensures accurate power sharing. In addition, the suggested controller only uses low-bandwidth communication (LBC). However, the adoption of too many PI controllers may compromise the system's stability. In [28], a multi-agent approach combines the voltage shifting term with double quadrant droop control, considering both real-time install capacity of battery and SoC value. This integration is aimed at achieving SoC equilibrium and DC bus voltage restoration. In Ref. [14], the droop coefficient adjusts dynamically based on SoC value changes, introducing an information parameter to improve power sharing and support DC bus voltage restoration. Within these distributed control studies, many methods achieve SoC balance through droop coefficient adjustments, yet this also affects system dynamics. In Ref. [29], the rate of SoC convergence was considered. The method involved incorporating an exponential function with the SoC to dynamically modify the droop coefficient. However, it's important to note that this approach shows reduced convergence speed as the SoC balancing process progresses. Therefore, a hybrid SoC equilibrium approach is developed in [30] to expedite the convergence speed during the subsequent phase of SoC equilibrium. Nevertheless, the application of this scheme necessitates the design of multiple convergence factors, and it brings about concerns regarding the system's limited stability margin. Moreover, Ref. [31] introduces a hierarchical cooperative control approach for parallel BSSs, achieving rapid SoC balancing through the adjustment of a single convergence factor. Nevertheless, this approach requires the coordination of numerous controllers to attain current distribution, SOC balance, and voltage regulation.

This study introduces an adaptive droop-based strategy for SoC equalization in parallel BSSs within a shipboard DC microgrid. This proposed methodology takes a comprehensive approach to address concerns such as DC bus voltage restoration, accurate load current distribution and SoC equilibrium. Table 1 presents a comparison between the proposed method and other SoC balancing control approaches for BSS in DC MG. When contrasted with existing methods in the same domain, this article provides the following principal contributions.

1. The controlling parameter SoC balance is realized through a sigmoid function. By modifying a single parameter within this algorithm, both the speed and accuracy of SoC equalization can be significantly improved.
2. The virtual voltage drops equalization (VVDE) controller, adaptive droop-based SoC (ADBS) controller and secondary compensation controller i.e.; bus voltage compensation (BVC) Observer are separately developed. These controllers can collaborate to achieve stable bus voltage, dynamic SoC equilibrium, and precise load current distribution.
3. An adjacent communication network is established utilizing a multi-agent consensus algorithm (MACA). Within this network, all BEUs achieve consensus through a protocol relying solely on local information as well as information from adjacent BSUs. This approach allows for the local estimation of pertinent average global variables.
4. Compared to conventional distributed control methods as implemented earlier, as demonstrated in Ref. [32], the presented approach considers factors like voltage drops, line impedance, and capacity difference. This method ensures accurate and fast SoC equilibrium throughout the process, which contrasts with scenarios in Ref. [32]. Moreover, it efficiently achieves accurate current sharing, fast SoC balancing, and steady voltage restoration using a limited count of controllers, differing from the approaches in literature. Extensive simulations and experiments validate the enhancements introduced by the proposed method, notably boosting the accuracy of SoC balancing and speed. Furthermore, this method not only withstands communication failures but also meets the system's dynamic demands even in instances of communication delays.

Table 1

Comparison of SoC balancing approaches for battery storage system (BSS) in DC MG.

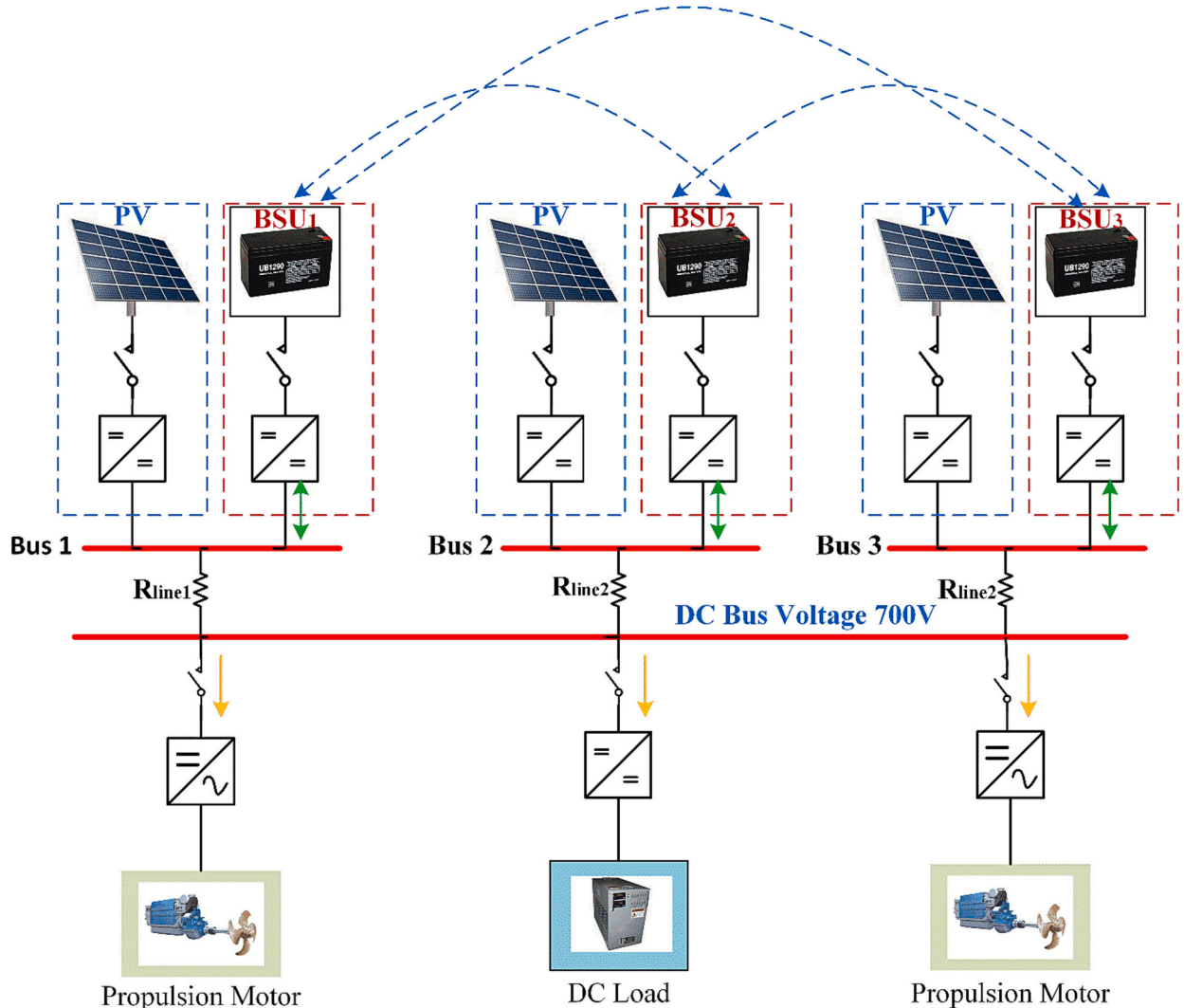
Literature scheme	SoC convergence factors	SoC balance speed	SoC balancing	SoC balancing accuracy	Capacity difference	Line impedance compensation	Communication topology	Voltage restoration
[23]	2	Slow	✓	Low	X	X	No	X
[24]	–	Slow	✓	Low	X	X	No	✓
[25]	–	Slow	✓	Low	X	X	No	✓
[26]	–	Slow	✓	Low	X	X	No	X
[27]	1	Slow	✓	Fast	X	✓	Neighbors	✓
[28]	1	Slow	✓	Fast	X	✓	Neighbors	✓
[14]	2	Fast	✓	Low	X	X	Neighbors	✓
[29]	1	Slow	✓	Low	X	X	All	X
[30]	3	Fast	✓	Low	✓	✓	All	✓
[31]	2	Slow	✓	Low	X	X	Neighbors	X
Proposed	1	Fast	✓	Fast	✓	✓	Neighbors	✓

2. System structure of DC shipboard microgrid and conventional droop control method

2.1. Structure of DC shipboard microgrid

The renewable energy-based microgrid system discussed in this paper is a microgrid system of a new electric propulsion ship, which consists of a battery storage system (BSS), daily load module, propulsion

system and multiple photovoltaic (PV) systems. Fig. 1 depicts the microgrid's typical structure. The PV system is connected through a power electronic converter to the DC bus, as distributed power supply relies on stable power generation. The BSS units are coupled through bidirectional DC/DC buck-boost converters to the DC bus. The main purpose of this system is to manage fluctuations in power due to PV generation and provide immediate support to the DC bus in exceptional scenarios. BSS primary purpose is to store extra energy produced by the

**Fig. 1.** Typical Structure of DC-SM.

PV array during the day so that it may be used to power loads at night or on days when it is raining. Therefore, BSS plays an important role in energy management and regulating bus voltage. In Fig. 1, the blue dashed arrows show the communication links among the BSUs, and the green and orange arrows show the direction of energy flow.

2.2. Traditional droop control approach

The conventional V-I droop control method is primarily utilized as the mechanism for load current distribution in DC MG. This control strategy is typically represented as follows:

$$U_{rj} = U_{bus}^* - I_{oj}R_{vj} \quad (1)$$

where U_{rj} represents the reference output voltage of BSU_j , R_{vj} and I_{oj} denote the droop coefficient and output current of BSU_j , U_{bus}^* is the rated DC bus voltage value, respectively.

Fig. 2 shows the equivalent model of a DC microgrid with two paralleled BSUs and according to Kirchhoff voltage law (KVL) and circuit analysis, it can be written as

$$\begin{cases} U_{PMC} = U_{o1}^* - I_{o1}(R_{v1} + R_{L1}) \\ U_{PMC} = U_{o2}^* - I_{o2}(R_{v2} + R_{L2}) \\ U_{PMC} = (I_{o1} + I_{o2})R_{Load} \end{cases} \quad (2)$$

where U_{PMC} is the voltage at point of mutual connection, representing the actual DC bus voltage value, R_{L1} , R_{L2} , and R_{Load} are the line impedance of BSU_1 and BSU_2 and load resistance.

It can be derived by combining (1) and (2),

$$\begin{cases} I_{o1} = \frac{AU_{o1}^* - R_{Load}U_{o2}^*}{C} \\ I_{o2} = \frac{BU_{o2}^* - R_{Load}U_{o1}^*}{C} \end{cases} \quad (3)$$

where as

$$A = (R_{v2} + R_{L2} + R_{Load}) \quad (4)$$

$$B = (R_{v1} + R_{L1} + R_{Load}) \quad (5)$$

$$C = (R_{v1} + R_{L1})(R_{v2} + R_{L2}) + (R_{v2} + R_{L2} + R_{v1} + R_{L1})R_{Load} \quad (6)$$

From (3) to (6), the ratio of output currents between the two BSUs can be calculated as

$$\frac{I_{o1}}{I_{o2}} = \begin{cases} \frac{R_{v2} + R_{L2}}{R_{v1} + R_{L1}}, & \text{if } U_{o1}^* = U_{o2}^* \\ \frac{AU_{o1}^* - R_{Load}U_{o2}^*}{BU_{o2}^* - R_{Load}U_{o1}^*}, & \text{if } U_{o1}^* \neq U_{o2}^* \\ \frac{R_{v2}}{R_{v1}}, & \text{if } U_{o1}^* = U_{o2}^*, R_{v1} \gg R_{L1}, R_{v2} \gg R_{L2} \end{cases} \quad (7)$$

From Eq. (7), it can be seen that when the voltage of both converters is the same, i.e., $U_{o1}^* = U_{o2}^*$, and the droop coefficient of both converters is much larger than the line impedance, the parallel converter's output current is inversely related to the corresponding virtual resistance. The temporary effect of the line impedance can be ignored. Therefore, the distribution of load power between converters can be effectively changed by adjusting the droop coefficients.

Furthermore, according to Eq. (1), by using the droop control, the converter's output voltage might deviate specified value and express the DC bus voltage's deviation value as

$$\Delta U_{oj} = I_{oj}R_{vj} \quad (8)$$

The droop coefficient is chosen considering the converter's maximum allowable output current and the acceptable range for bus voltage deviation. Choosing a droop coefficient that is excessively large or too small can impact the stability of the system. Thus, the value is constrained within specific limits to ensure system stability.

$$0 < R_{vj} \leq \frac{\Delta U_{bus \max}}{I_{omax}} \quad (9)$$

Here, $\Delta U_{bus \max}$ represents the maximum variation in the bus voltage, while I_{omax} stands for the converter's maximum acceptable output current.

3. Proposed control strategy

This article presents the comprehensive schematic diagram of the introduced control method in Fig. 3. Each unit is linked in parallel to the microgrid bus, and relevant mean data calculated by neighboring BSUs, is communicated through an LBC. The drive signals for the switching tubes are labeled as Q_1 and Q_2 . R_{Lj} represents the line impedance that forms a connection between the converter and the microgrid bus. $i_{oj}(t)$, $i_{Lj}(t)$ and $U_{oj}(t)$ represent the output current, inductor current, and real-time local measurement of output voltage, respectively. Additionally, the control layer's parameters would be introduced below. Also, a BSS hierarchical control structure is implemented, primarily comprising the communication, primary control, and secondary control layer.

In the primary control layer, an adaptive droop-based SOC controller

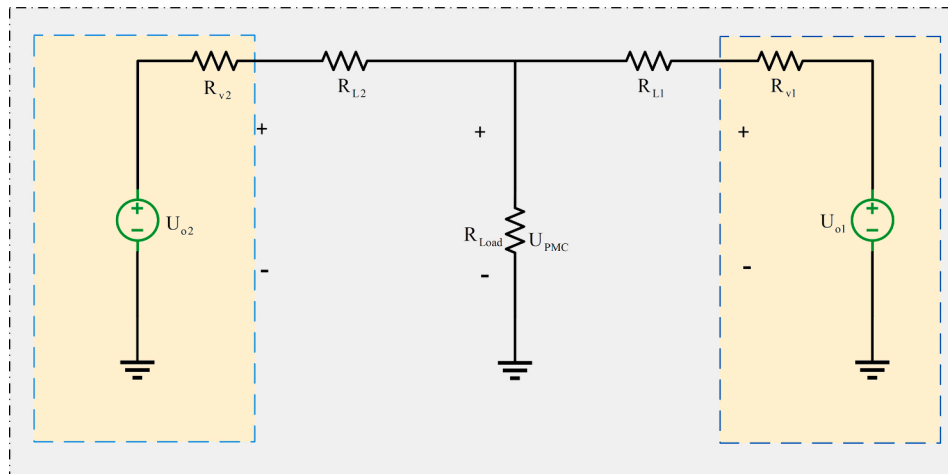


Fig. 2. Equivalent circuit schematic diagram.

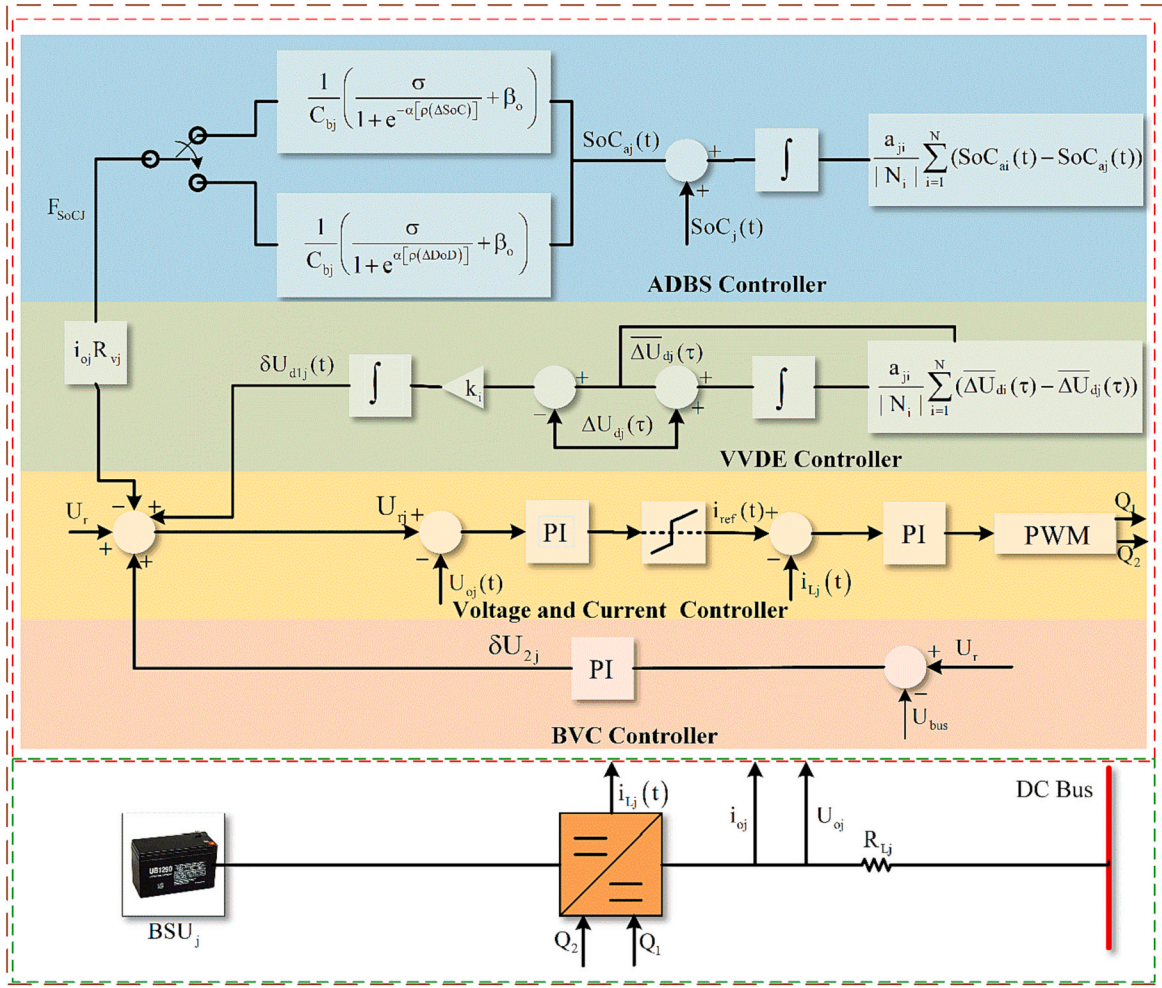


Fig. 3. Proposed model control diagram.

(ADBS) is designed based on SoC. Its purpose is to achieve dynamic SoC equilibrium while ensuring equitable sharing of load current. Within the secondary control layer, there are two controllers: The bus voltage compensation (BVC) and the virtual voltage drops equalization (VVDE) controller. These controllers are designed to mitigate the impact of disparate line impedance and rectify deviations in DC bus voltage resulting from the interplay between the line impedance and droop coefficient. Within the communication layer, a network of peer-to-peer communication is established using a multi-agent approach. In this arrangement, each BSU engages exclusively with its adjacent BSUs. Through an iterative multi-agent consensus algorithm, relevant average global variables are estimated, driving them to collectively converge towards the intended values. Fig. 4 represents the communication topology among the distributed BSUs.

3.1. Multiagent agent consensus algorithm (MACA)

Within multiagent systems, the consensus algorithm maintains consistent agent states without central control or global communication [33]. The power network's evolution towards distributed intelligent cooperation designates each DC microgrid BSU as an agent adhering to these protocols. Through iterative communication with neighboring BSUs, global average variables converge. Distributed control, distinct from centralized methods, provides global status monitoring, robustness in emergencies (e.g., disconnections, attacks), and advantages like low communication costs and scalability.

The DC microgrid's communication network, shown in Fig. 4,

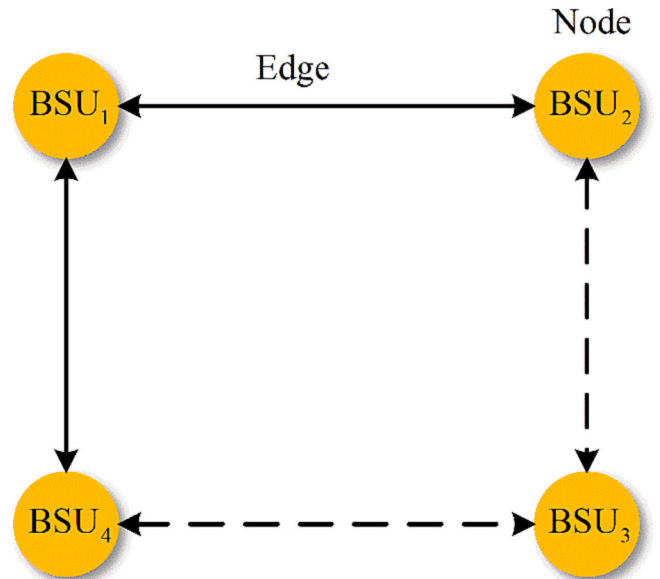


Fig. 4. Communication topology.

involves N BSUs nodes forming an undirected connectivity graph $G_N = (U, E, M)$. Here, E stands for the edge set among BSUs and U represents $\{BSU_1, BSU_2, BSU_3, \dots, BSU_N\}$, signifying the node set. The adjacency

of $(BSU_i, BSU_j) \in E$ denotes nodes capable of data transmission. The matrix $M = [a_{ij}] \in R^{N \times N}$ characterizes node-edge relationships, with a_{ij} representing communication weights. For $(BSU_i, BSU_j) \in E$, $a_{ij} > 0$; otherwise, $a_{ij} = 0$. The Laplacian matrix $L = [l_{ij}] \in R^{N \times N}$ can be represented as

$$\begin{cases} l_{ij} = \sum_{j=1, j \neq i}^N a_{ij} \\ l_{ij} = -a_{ij}, j \neq i \end{cases} \quad (10)$$

A well establish multi-agent consensus algorithm is proposed in [34]

$$\dot{x}_i(t) = - \sum_{j \in N_i} a_{ij} (x_i(t) - x_j(t)) + \dot{\gamma}_i(t) \quad (11)$$

$x_i(t)$ and $x_j(t)$ represent the node j and i state variables, respectively, while $\dot{\gamma}_i(t)$ representing a local input variable

Subsequently, the global dynamics of the multi-agent consensus algorithm can be expressed as:

$$\dot{x}(t) = -Lx(t) + \dot{\gamma}(t) \quad (12)$$

According to eqs. (11) and (12), if a communication failure occurs and the graph G_N includes at least one directed spanning tree, then eq. (12) can meet the requirement of approaching $\lim_{t \rightarrow \infty} |x_i(t) - x_j(t)| = 0$. Hence, the article selects a cost-effective ring topology for BSU communication, ensuring variable convergence during communication failure.

3.2. Adaptive droop based SoC controller

The SoC has a critical impact on the battery's lifespan. If the SoC surpasses predefined limits, subjecting the battery to overcharging or deep discharge may result in deterioration of the batteries lifespan. Therefore, to prevent the shortened service life of specific BSUs, achieving SoC balance is essential. This equilibrium ensures that each BSU allocates its power reasonably, based on its individual SoC level

As obtaining SoC directly from the battery itself is not feasible, the precise estimation of battery SoC becomes a critical task. In this context, the fundamental technique known as coulomb counting [35] is adopted to estimate the SoC for each battery, as calculated below:

$$SoC_j(t) = SoC_j(0) - \frac{\mu_j}{C_{bj}} \int_0^t i_{batj}(\tau) d\tau \quad (13)$$

where $SoC_j(0)$ and $SoC_j(t)$ represent the initial and current SoC values at time for the BSU_j . i_{batj} , C_{bj} and μ_j represent the output current, rated capacity of BSU_j , and charge/discharge efficiency of the BSU_j .

The rate of change of SoC can be determined from (13)

$$SoC'_j(t) = - \frac{\mu_j \cdot i_{batj}}{C_{bj}} \quad (14)$$

when the μ_j of the two converters is identical, the ratio of the derivatives of SoC for the two BSUs under typical droop control can be expressed as follows:

$$\frac{SoC'_1(t)}{SoC'_2(t)} = \frac{C_{b2}(R_{v2} + R_{L2})}{C_{b1}(R_{v1} + R_{L1})} \quad (15)$$

Derived from Eq. (15), the balance of SoC is interrelated with the variables R_{Lj} , C_{bj} , and R_{vj} . Whereas the parameters, C_{bj} and R_{Lj} cannot be modified arbitrarily. To attain SoC equilibrium, a viable approach is to establish a functional relationship between the droop coefficient F_{SoCj} and the SoC. This coefficient is dynamically adjusted based on the evolving SoC value, thereby facilitating the achievement of the desired SoC equalization objective.

Therefore, to construct the F_{SoCj} , a sigmoid function $\frac{1}{1+e^{-x}}$ based on

the SoC is presented in this paper, which is expressed as follows:

$$F_{SoCj} = \begin{cases} \frac{1}{C_{bj}} \left(\frac{\sigma}{1 + e^{-\alpha[\rho(\Delta SoC)]}} + \beta_o \right) i_{oj} < 0 \\ \frac{1}{C_{bj}} \left(\frac{\sigma}{1 + e^{\alpha[\rho(\Delta SoD)]}} + \beta_o \right) i_{oj} > 0 \end{cases} \quad (16)$$

where,

$$\rho = \begin{cases} \frac{1}{|\Delta DoD| + \varepsilon} \text{ discharging} \\ \frac{1}{|\Delta SoC| + \varepsilon} \text{ charging} \end{cases} \quad (17)$$

$$\Delta SoC = SoC_j - SoC_a \quad (18)$$

$$\Delta DoD = DoD_a - DoD_j \quad (19)$$

From Eq. (16), it is apparent that the factor F_{SoCj} (where $F_{SoCj} > 0$) is subject to adaptive adjustments based on the SoC value. Consequently, the algorithm exhibits two distinct behaviors: $i_{oj} < 0$ indicates BSU charging, $i_{oj} > 0$ indicates BSU discharging. The signs σ and ε represent the adjustment parameters for balance speed and accuracy, respectively. These factors have a significant impact in controlling the speed and accuracy of SoC balance. β_o represents the initial droop coefficient. According to Eq. (17), ρ is the acceleration factor.

From (18) and (19), DoD_j represents the discharge depth of the j -th converter, with its value calculated as $DoD_j = 1 - SoC_j$. DoD_a and SoC_a denote the average discharge depth and average SoC, respectively. According to Eq. (16), in the charging process (where $i_{oj} < 0$) when $DoD_j > DoD_a$ the droop coefficient becomes larger, causing the charging current to decrease. This results in a slower increase in SoC. Conversely, when $DoD_j < DoD_a$, the droop coefficient is small. As a result, the charging current increases, leading to a faster rise in SoC. Similarly, during the discharging process (where $i_{oj} > 0$) when $SoC_j > SoC_a$ the droop coefficient decreases. This leads to an augmented discharging current, causing the state of charge to decrease more rapidly. Conversely, $SoC_j < SoC_a$ the droop coefficient increases. Consequently, the discharging current diminishes, resulting in a slower decline of the state of charge. Based on this rule, the charging/discharging process facilitates a gradual convergence of the SoC. This gradual approach effectively mitigates the possibilities of overcharging and excessive discharge in specific BSUs, consequently enhancing their operational lifespan

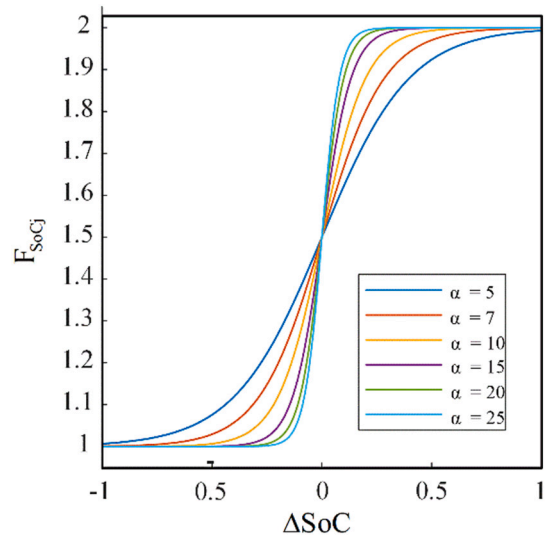


Fig. 5. The F_{SoCj} curve showing ΔSoC variations for various α values against the surface graph.

significantly.

Fig. 5 illustrates the variations in the F_{SoCj} curve with ΔSOC across different α values. Even, when the change in ΔSOC approaches zero, the F_{SoCj} values remain notably elevated after applying the sigmoid function. This retention of significant F_{SoCj} values via sigmoid transformation ensures the reliable achievement of SoC convergence. Fig. 5 concluded that during BSU discharging ($\Delta SOC < 0$), the BSU has less stored energy, this leads to an increased droop coefficient and a decreased discharge current as a result. At same time, if $\Delta SOC > 0$, the BSU has more energy stored, hence it decreased the droop coefficient and discharge current should be increased. In conclusion, equilibrium is reached for the SoC of each BSUs, ensuring accurate distribution of the load power. The analysis presented above underscores the logical foundation of the ADBS controller's approach. Similarly, a comparable deduction can be applied to ascertain the charging conditions.

A qualitative evaluation of parameter selection involving ε , α , and σ is enriched by the inclusion of visual representations. It is evident that the equilibrium function F_{SoCj} is chosen to be a sigmoid function due to its strong symmetry, distinct asymptotes of -1 and 1 , and its assurance of maintaining the fundamental SoC equilibrium trend. Therefore, β_o is employed to shift F_{SoCj} to ensure it remains above 0, ensuring the stability of the system. The surface graph presented in Fig. 6(a) illustrates the behavior of the acceleration factor ρ in response to changes in the parameters ε and ΔSoC . Notably, when the value of ε is < 0.03 , the ρ exhibits a significant increase in gain as ΔSoC approaches proximity to 0. This specific behavior results in a heightened rate of change for the equilibrium function F_{SoCj} during this distinct phase. The accelerated rate of change effectively expedites the equilibrium adjustment process, particularly in the subsequent stages. Conversely, in situations where ΔSoC significantly deviates from 0, the regulatory factor ρ must be adjusted to deliver a gain value nearly equal to 1, ensuring the stability of the SoC equilibrium is maintained. Given these factors, a reasonable choice would be to assign a value of 0.01 to the parameter ε .

Fig. 6(b) and (c) provides an explaining surface diagram representing the equilibrium function F_{SoCj} in response to varying parameters α and σ . Notably, parameter σ significantly impacts the upper boundary of F_{SoCj} , while parameter α plays a key role in shaping the fundamental trajectory and pace of SoC equilibrium. To achieve a balanced range of F_{SoCj} that maintains both moderation and equilibrium, a favorable selection for parameter σ is proposed at a value of 3.5. However, it's important to highlight that when parameter α is below 3, the observed change in F_{SoCj} as ΔSoC gradually deviates from 0 deviates from the earlier analysis, resulting in an unsatisfactory outcome for SoC

equilibrium. As parameter α gradually increases from 3, there's a notable improvement in the consistency of equilibrium results, and the equilibrium speed progressively rises. Taking into account, the overall equilibrium speed of the process, a value of > 3 for parameter α is deemed suitable. These considerations extend to both discharge and charging scenarios, demonstrating the qualitative analysis of parameter selection for ε , α , and σ in a comprehensive manner. From the previously discussed analysis results, it is apparent that σ primarily influences the speed of the initial stage of SoC equilibrium, whereas ε predominantly affects the accuracy of the final stage of SoC balancing. Therefore, choosing suitable values for σ and ε in accordance with specific requirements is essential to achieve a more favorable balance between SoC balancing speed and accuracy.

To address the challenge of system communication overload, an effective MACA is used in this article which can estimate SoC_a and DoD_a to develop an average SoC and DoD observers. This observer utilizes data available locally and information obtained from neighboring elements to continually update the locally calculated average SoC and DoD. By combining this concept with Eq. (11), the average SoC and DoD observers are formulated as:

For average SoC

$$SoC_{aj}(t) = SoC_j(t) + \frac{a_{ji}}{|N_j|} \int_0^t \sum_{i \in N_j} (SoC_{ai}(\tau) - SoC_{aj}(\tau)) d\tau \quad (20)$$

For average DoD

$$DoD_{aj}(t) = DoD_j(t) + \frac{a_{ji}}{|N_j|} \int_0^t \sum_{i \in N_j} (DoD_{ai}(\tau) - DoD_{aj}(\tau)) d\tau \quad (21)$$

Here, $SoC_{aj}(t)$ and $DoD_{aj}(t)$ represent the average observed SoC and DoD at time (t) for the BSU_j , while $SoC_j(t)$ and $DoD_j(t)$ denote the SoC and DoD measured at time (t) for the BSU_j . The parameter a_{ji} signifies the average SoC and DoD observers gain. Additionally, $SoC_{ai}(\tau)$, $DoD_{aj}(\tau)$ and $DoD_{ai}(\tau)$ correspond to the average SoC and DoD observed at time (τ) for the BSU_j and BSU_i , respectively.

3.3. Virtual voltage drops equalization (VVDE)

The SoC balance controller ensures balanced load current, but variations in line impedance among BSUs converters lead to imprecise current sharing. To address this issue, a VVDE observer is introduced. This observer calculates average voltage drop $\Delta \bar{U}_{df}(t)$ and local voltage

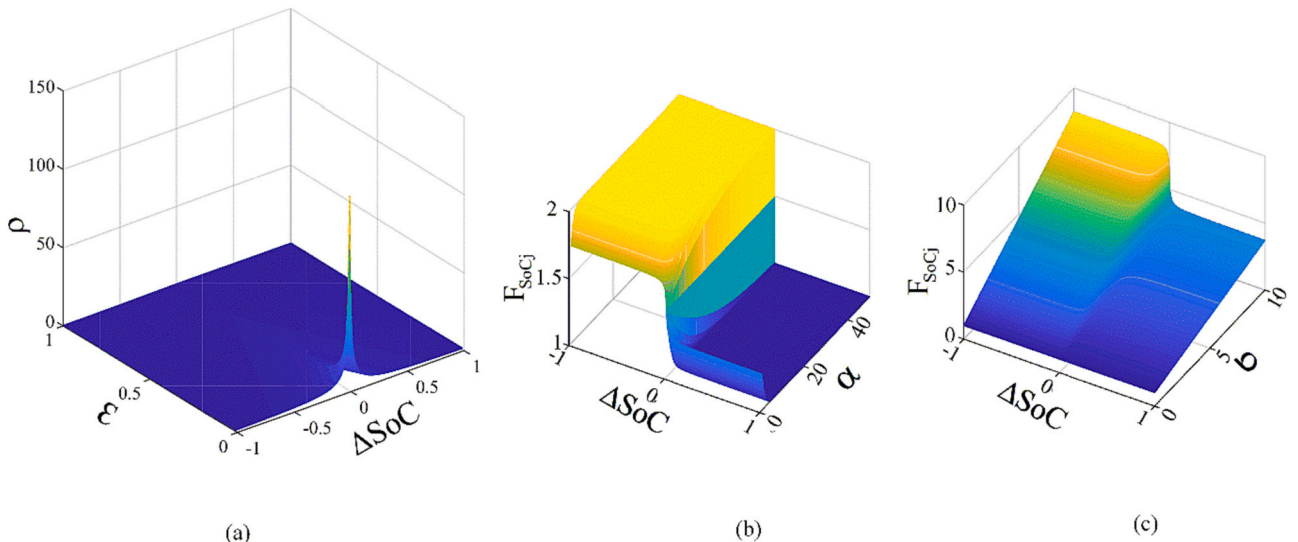


Fig. 6. F_{SoCj} surface graph trend with parameters ε , α , σ and ΔSoC variations.

drop $\Delta U_{dj}(t) = i_{oj}(t)R_{vj}(t)$. Through an integrator-based adjustment of the difference between these variables, a compensation term δU_{dlj} is generated. This compensation is then added to the output reference voltage, indirectly rectifying current distribution errors due to uneven line impedance.

From Eq. (22), each BSU estimates $\overline{\Delta U_{dj}}(t)$ locally through an average VVDE, as expressed below.

$$\overline{\Delta U_{dj}}(t) = \Delta U_{dj}(t) + \frac{a_{ji}}{|N_j|} \int_0^t \sum_{i \in N_j} (\overline{\Delta U_{di}}(\tau) - \overline{\Delta U_{dj}}(\tau)) d\tau \quad (22)$$

where a_j parameter denote the gain of VVDE controller

The distinction between $\overline{\Delta U_{dj}}(t)$ and $\Delta U_{dj}(t)$ is adjusted through integration, resulting in the voltage correction term which can be expressed as:

$$\delta U_{dlj} = K_i \int_0^t (\overline{\Delta U_{di}}(\tau) - \Delta U_{dj}(t)) \quad (23)$$

where parameter K_i represent the integrator gain.

3.4. Recovery strategy for bus voltage

3.4.1. Bus voltage compensation (BVC) controller

The proposed ADBS and VVDE observer ensure fast SoC balancing and accurately distribute the power to each BSU. However, applying droop control to parallel converters can cause bus voltage deviations, where the voltage rises during charging and drops during discharging. To overcome this, a recovery strategy for DC bus voltage is introduced in this article, which stabilizes the voltage nearby the target value. This method is effective during regular operation and in the event of communication failure scenarios and requires a simple PI controller for implementation. The formula for bus voltage compensation can be formulated as follows:

$$\delta U_{2j} = \left(K_p + \frac{K_i}{s} \right) (U_r - U_{bus}) \quad (24)$$

Here, δU_{2j} signifies the secondary compensation for DC bus voltage, while parameters K_p and K_i represent the proportional and integral coefficients of the PI controller within the compensation loop of the DC bus voltage, respectively. Therefore, the finalized enhanced droop control term can be expressed as:

$$U_{rj} = U_{bus}^* - I_{oj}R_{vj} F_{SoCj} + \delta U_{dlj} + \delta U_{2j} \quad (25)$$

4. Analysis of the proposed system's stability

In this section, to analyse the stability of the DC microgrid, the equivalent circuit of two parallel BSUs depicted in Fig. 2 is taken as an example and output current of two parallel converters can be express as

$$i_{o1} = \Phi_1 U_{o1} - \lambda U_{o2} \quad (26)$$

$$i_{o2} = \Phi_2 U_{o2} - \lambda U_{o1} \quad (27)$$

where,

$$\begin{cases} \Phi_1 = \frac{R_{Load} + R_{L2}}{R_{L1}R_{L2} + R_{Load}R_{L1} + R_{Load}R_{L2}} \\ \Phi_2 = \frac{R_{Load} + R_{L1}}{R_{L1}R_{L2} + R_{Load}R_{L1} + R_{Load}R_{L2}} \\ \lambda = \frac{R_{Load}}{R_{L1}R_{L2} + R_{Load}R_{L1} + R_{Load}R_{L2}} \end{cases} \quad (28)$$

In Fig. 7, the simplified model for stability analysis is displayed and SoC estimation transfer function can be calculated as:

$$G_{est}(s) = \frac{1}{s+1} \quad (29)$$

The closed-loop transfer function concerning DC voltage can be simplified as follows.

$$G_c(s) = \frac{1}{\tau s + 1} \approx 1 \quad (30)$$

Here, τ denotes the time constant influenced by the switch's operation.

The output current is associated with a low-pass filter (LPF), and the associated transfer function is as follows.

$$G_{lpf}(s) = \frac{\omega_c}{s + \omega_c} \quad (31)$$

Here, ω_c represents the cut-off frequency of the low-pass filter.

Based on the information presented in Fig. 9 for converter 1, and Eqs. (27) to (32), the subsequent relationship can be express as:

$$U_{o1} = U_r - G_{est}(s)G_{lpf}(s)R_{v1}(\Phi_1 U_{o1} - \lambda U_{o2}) \quad (32)$$

Without considering the communication lag between BSUs, by combining (27)–(33) the output voltage of converter 1 can be calculated as

$$U_{o1} = \frac{U_r G_c(s) - G_{est}(s)G_c(s)G_{lpf}(s)R_{v1}\lambda U_{o2}}{1 + G_{est}(s)G_c(s)G_{lpf}(s)R_{v1}\Phi_1} \quad (33)$$

At $U_{o2} = 0$

$$\frac{U_{o1}}{U_r} = \frac{G_c(s)}{1 + G_{est}(s)G_c(s)G_{lpf}(s)R_{v1}\Phi_1} \quad (34)$$

So, system characteristic equation can be obtain based on Eq. (35)

$$Ds^2 + Es + F = 0 \quad (35)$$

where

$$D = 1$$

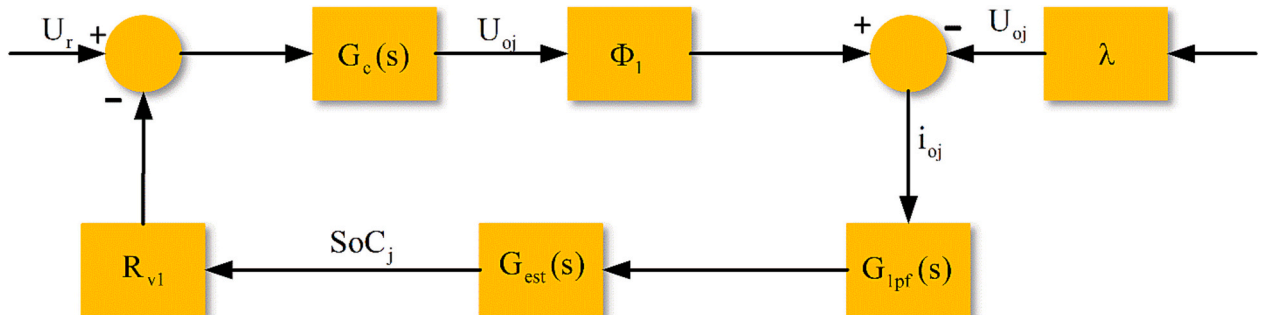


Fig. 7. Simplified stability analysis model.

$$E = (1 + \omega_c)$$

$$F = \omega_c(1 + R_{v1}\Phi_1)$$

Fig. 8 presents a visualization of the root locus analysis for F_{SoCj} , and ω_c parameters. By examining Fig. 8, it becomes evident that the stability of the system is significantly affected by two main factors: the cut-off frequency ω_c and the droop coefficient F_{SoCj} . It's important to avoid setting F_{SoCj} at an excessively high value. Instead, the ideal value should be determined by considering Eq. (9) along with the results of stability analysis. Additionally, it's crucial to strike a balance between achieving accurate current sharing and minimizing bus voltage deviations. When it comes to choosing the cut-off frequency ω_c , the challenge involves managing the trade-off between overshoot and the time required for the system to settle. This decision is primarily influenced by the sampling frequency and the inherent characteristics of the system's hardware [36].

5. Proposed method simulation results

To assess the viability and effectiveness of our proposed control approach, a DC microgrid has been established in MATLAB/Simulink, incorporating three battery storage units (BSUs), DC/DC buck boost converters, photovoltaic (PV) system and controllable load. The structure of the simulation model is illustrated in Fig. 9, and we have designed six distinct simulation scenarios to test the performance of the suggested control approach. Furthermore, we have conducted a comparative analysis between the results of one case of our control strategy, as outlined in this article, and the approach presented in [32]. A comprehensive set of simulation parameters is provided in Table 2.

5.1. Case study I: evaluation of the proposed strategy during normal charge and discharge processes

This scenario primarily examines the control performance of the proposed approach under steady charging and discharging conditions of the BSUs under the normal condition of system. For the charging phase, the initial value of SoCs for the three BSUs is set at 25 %, 20 %, and 15 % respectively, while the output power of PV consistently generates 30 kW, and the controllable load power remains at 10 kW. In the discharging phase, the initial value of SoC for each BSU is set at 80 %, 75 %, and 70 %, respectively. The PV system output power is configured at 5 kW, with the load power remaining at 20 kW. Furthermore, throughout both the charging and discharging phases, the rated capacity is

consistently set at 50 Ah for all three BSUs.

Fig. 10 presents simulation results for stable charging and discharging processes of the BSUs, with (a), (b), and (c) showing charging phase results and (d), (e), and (f) illustrating discharge phase results. In Fig. 10(a), (b), (d), and (e), it becomes evident that the implementation of the approach proposed in this paper results in the stable convergence of the SoC for each BSU, along with a gradual alignment of the output currents among the converters. At approximately $t = 17$ s, we observe the achievement of SoC balance and synchronized current sharing among all converters, demonstrating a relatively high level of speed and precision in the current-sharing process. Moreover, as seen in Fig. 10(c), and (d), during the regular charging and discharging processes, the method presented in this paper demonstrates the ability to maintain the bus voltage at approximately 700 V.

The simulation outcomes validate that the proposed approach can rapidly and consistently achieve SoC convergence throughout both the charging and discharging operations of the BSS. Additionally, it ensures an accurate sharing of load currents and maintains the bus voltage within acceptable limits.

5.2. Case study II: evaluation of the proposed strategy under simultaneous sudden load and PV power changes

This scenario primarily concerns the effectiveness of the proposed strategy's behavior when there are abrupt fluctuations in either output power of PV system or load power, while the system operates under normal communication. Due to variations in the aging levels of the BSUs, the three BSUs' initial SoC values are 50 %, 45 %, and 40 %, respectively and each BSU possesses a rated capacity of 9 Ah. The simulation process is categorized into the following five phases.

Phase I (0 to 10s): The initial PV power is 20 kW, and load power is set at 15 kW.

Phase II (10 to 20s): PV power is set the same as in phase I, and then the load power is suddenly changed to 30 kW.

Phase III (20 to 30s): PV power is same as in phase I and II, and load power is changed from 30kW to 18 kW.

Phase IV (30 to 40s): In this phase, PV power and load power change at the same time, with PV power set at 12 kW and load power at 30 kW, respectively.

Phase V (40 to 50s): PV power is 42 kW, and the load power is decreased from 30 kW to 15 kW.

Fig. 11 illustrates the simulation results in response to abrupt changes in both output power of PV system and load power. At phase I ($t = 0$ to 10s), the output power generated by the PV system exceeds the

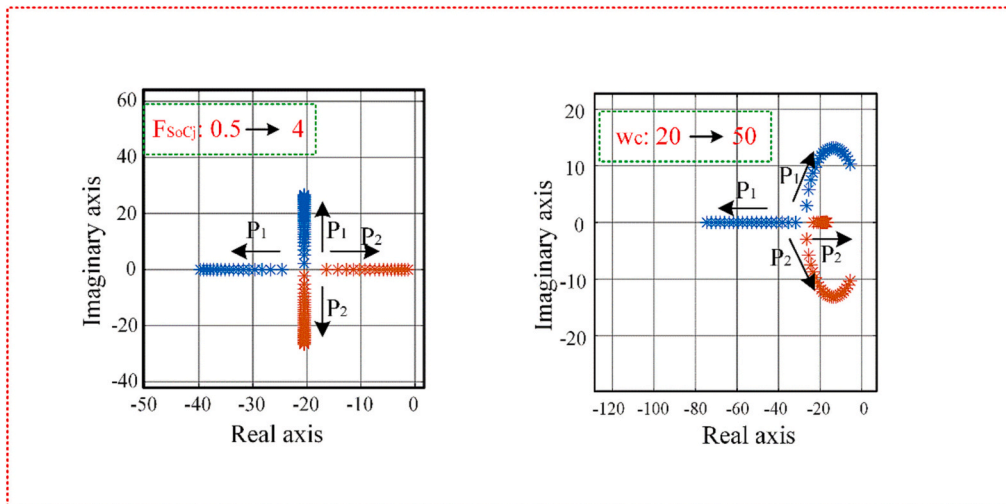


Fig. 8. Visualization of the root locus analysis of parameters F_{SoCj} , and ω_c .

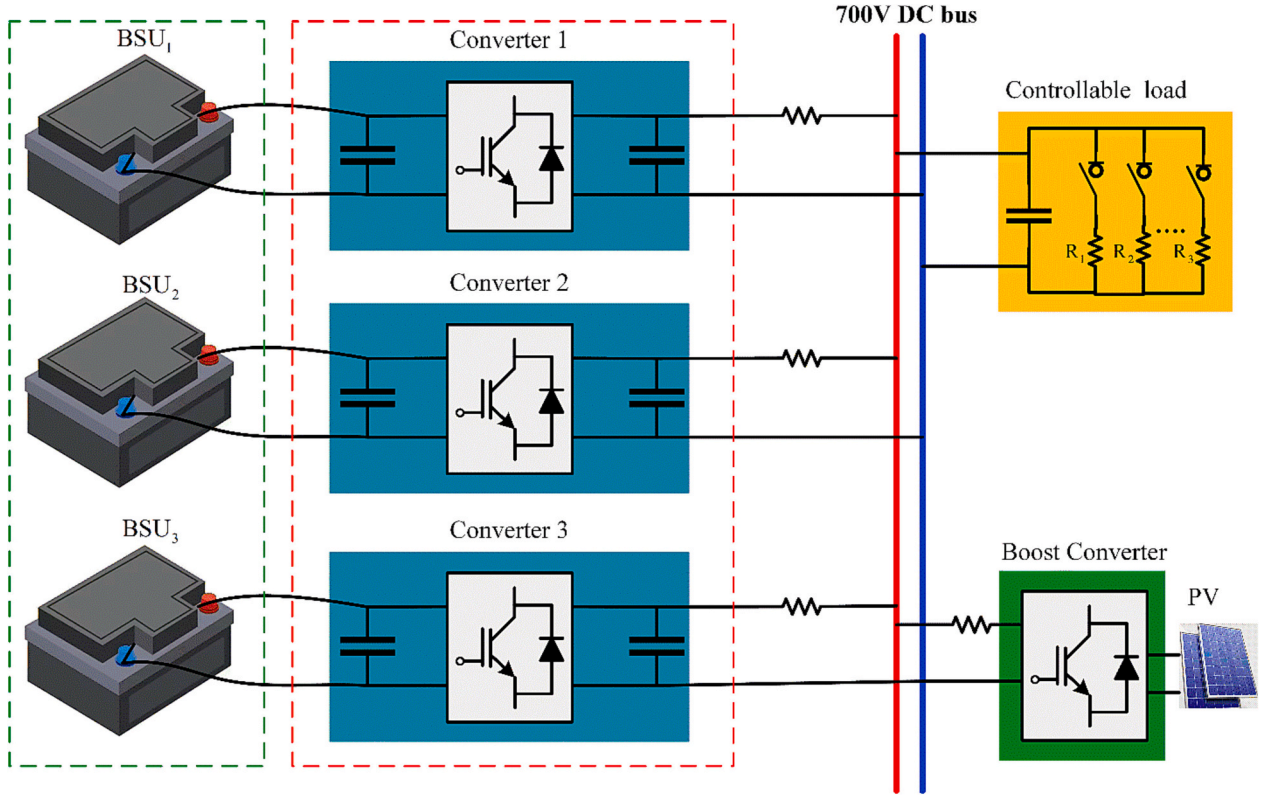


Fig. 9. Structure of simulation model.

Table 2
Parameter of simulation model.

Parameter	Symbol	Value
Rated DC bus voltage	$U_{bus}(V)$	700
Three BSU nominal voltage	$U_{o1}(V)$	700
	$U_{o2}(V)$	700
	$U_{o3}(V)$	700
Three BSU converters line impedance	$R_{L1}(\Omega)$	0.5
	$R_{L2}(\Omega)$	0.8
	$R_{L3}(\Omega)$	1.5
	$L(mH)$	2
DC/DC converter inductance	$C(nH)$	5
DC/DC converter capacitance	K_{PI}, K_{II}	2,60
PI controller current loop	K_{PV}, K_{IV}	1,40
PI controller voltage loop	K_{PU}, K_{IU}	8,80
Bus voltage controller	R_V	1
Droop coefficient	α	1
Convergence factor	$\epsilon, \sigma, \beta_0$	0.01, 3, 3.5
Speed and Accuracy factors		

load consumption, leading each BSU to operate in charging mode to absorb the excess power. However, at $t = 10$ s, the PV array's output becomes insufficient to meet the load, causing each BSU to transition from charging mode to discharging mode to compensate for the shortfall. At $t = 20$ s, there is a sudden drop in the load power to 18 kW, prompting each BSU to transition back to charging mode to maintain the system power balance. During this timeframe, there is a gradual convergence of both the SoC and load current. At $t = 30$ s, the output power of PV system suddenly decreases to 12 kW, while suddenly it changes the load from 18 kW to 30 kW. In response, each BSU operates in discharging mode to balance the system power shortage. At $t = 40$ s, the output power generated by the PV system suddenly increases to 42 kW, while the load decreases to 15 kW. Each BSU switches back to charging mode to store the excess power. Throughout the entirety of the simulation timeframe, the approach presented in this article effectively achieves SoC equilibrium and accurate load current distribution within

the 20 to 30 s timeframe, approximately at $t = 23$ s. In scenarios where load power, PV power, or both undergo rapid changes, fluctuations in bus voltage are observed. However, the proposed method effectively restores the bus voltage to 700 V. Additionally, both voltage and current exhibit improved dynamic performance during transient power fluctuations.

The aforementioned analysis confirms that the proposed scheme achieves faster equalization speed and higher accuracy. The dynamic effect of charge and discharge switching for each BSU is excellent, and it can and maintains the bus voltage within acceptable limits.

5.3. Case study III: evaluation of the proposed strategy with communication failure and different capacity

This case study mainly focuses on the effectiveness of the proposed strategy's behavior when each BSU has a different rated capacity in situations involving communication failures. The initial SoC for each BSU are 30 %, 35 % and 40 % respectively and their rated capacities are 1 Ah, 3 Ah, and 5 Ah, respectively. The time frame of simulation process is divided into three phases.

Phase I (0 to 15 s): The output power of PV system is set at 3.1 kW, and load power is 5 kW. During this phase, communication between BSU₁ and BSU₂ failed within the time interval from 0 to 6 s.

Phase II (15 to 30 s): The output power of PV is 5.1 kW, while the total load power is 1 kW.

Phase III (30 to 50 s): The output power of PV system decreases from 5.1 kW to 3 kW, while the load power increases from 1 kW to 4 kW.

The simulation results for case study III are shown in Fig. 12(a), (b), and (c). It is evident that the performance of the suggested strategy in this article remains unaffected even in the case of communication failure between BSU₁ and BSU₂. This resilience is attributed to the continuous existence of the minimum spanning tree within the communication system, which enables MACA to consistently estimate the average values of relevant variables. At $t = 27$ s around, the system has achieved SoC

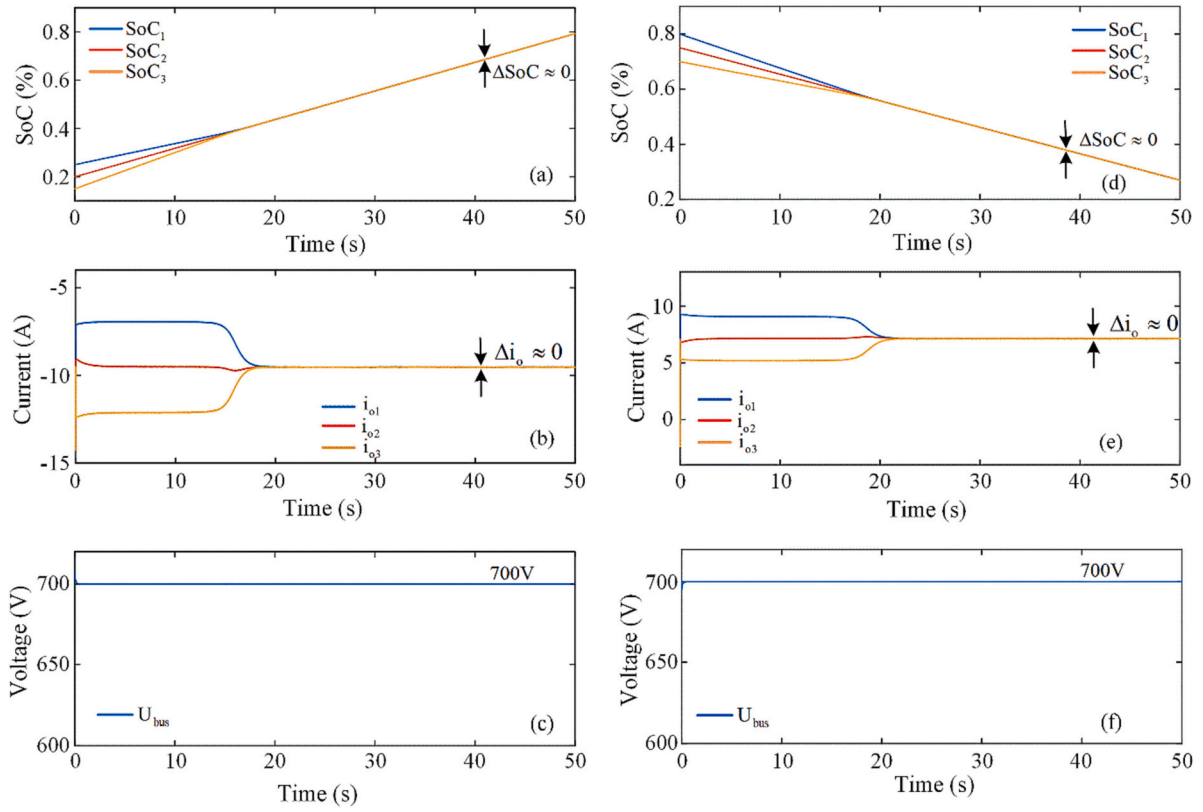


Fig. 10. Simulation result (a), (b), (c) of stable charging and (d), (e), (f) of discharging.

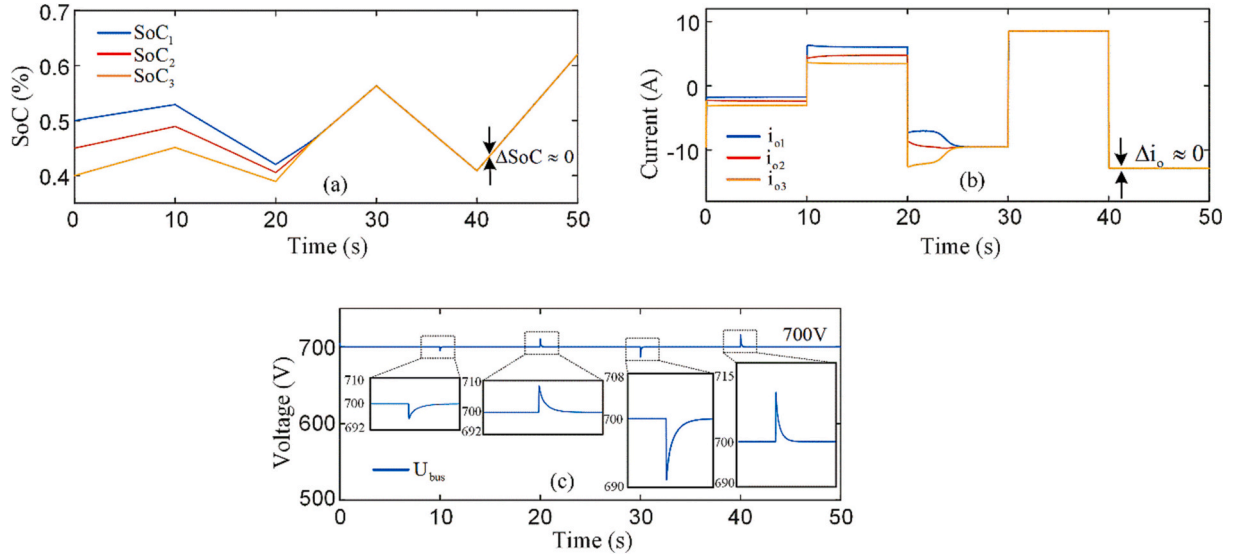


Fig. 11. Simulation result of PV fluctuation and sudden load changes.

balance and proportional load current distribution. Furthermore, the output current ratio is maintained at 1:3:5, with the SoC balance error approximately zero. Finally, the simulation result of scenario III validates the performance of the proposed method in addressing the sudden failure issue.

The above analysis confirms that the proposed approach can still maintain normal system operation and ensure the continuous convergence of the SoC for each BSU and proportional load current distribution, even in the presence of a communication failure, with the different rated capacity of each BSU.

5.4. Case study IV: comparison with advance method

In this case, we conduct a comparative study between the proposed method and the method presented in Ref. [32] under conditions of load and PV fluctuations. The approach described in Ref. [32] does not account for line impedance compensation, or the restoration of bus voltage. Consequently, it demonstrates a slower rate of SoC balance and lower accuracy in load current sharing. To validate the capability of the proposed control approach in addressing these concerns and to emphasize its advantages, this scenario draws a comparison between the

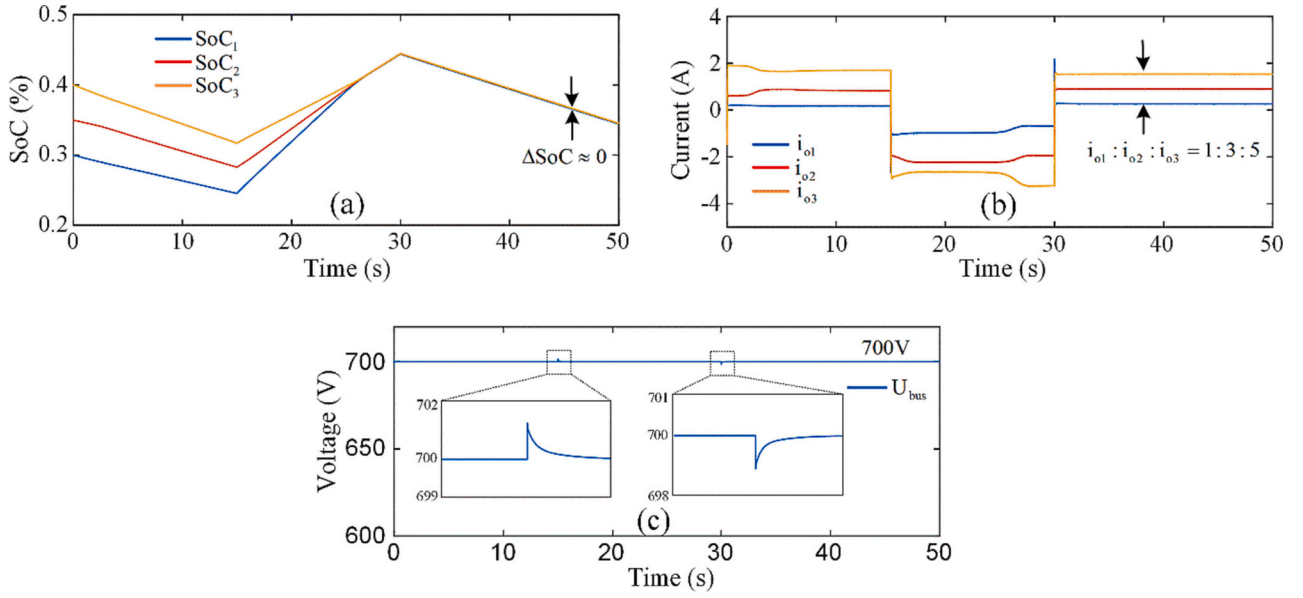


Fig. 12. Simulation result of communication failure and different capacity.

proposed approach and the strategy described in Ref [32]. The initial SoC for each BSU are 40 %, 35 %, and 30 %, respectively, while the capacity for each BSU is set at 7 Ah. Additionally, when simulating the SoC equalization approach described in Ref. [32], we set the parameter

p to 4. The simulation process is categorized into five phases.

Phase I (0 to 10s): The initial PV power and load power are set at 25 kW and 15 kW, respectively. The proposed model and the reference model are activated at $t = 0$ s.

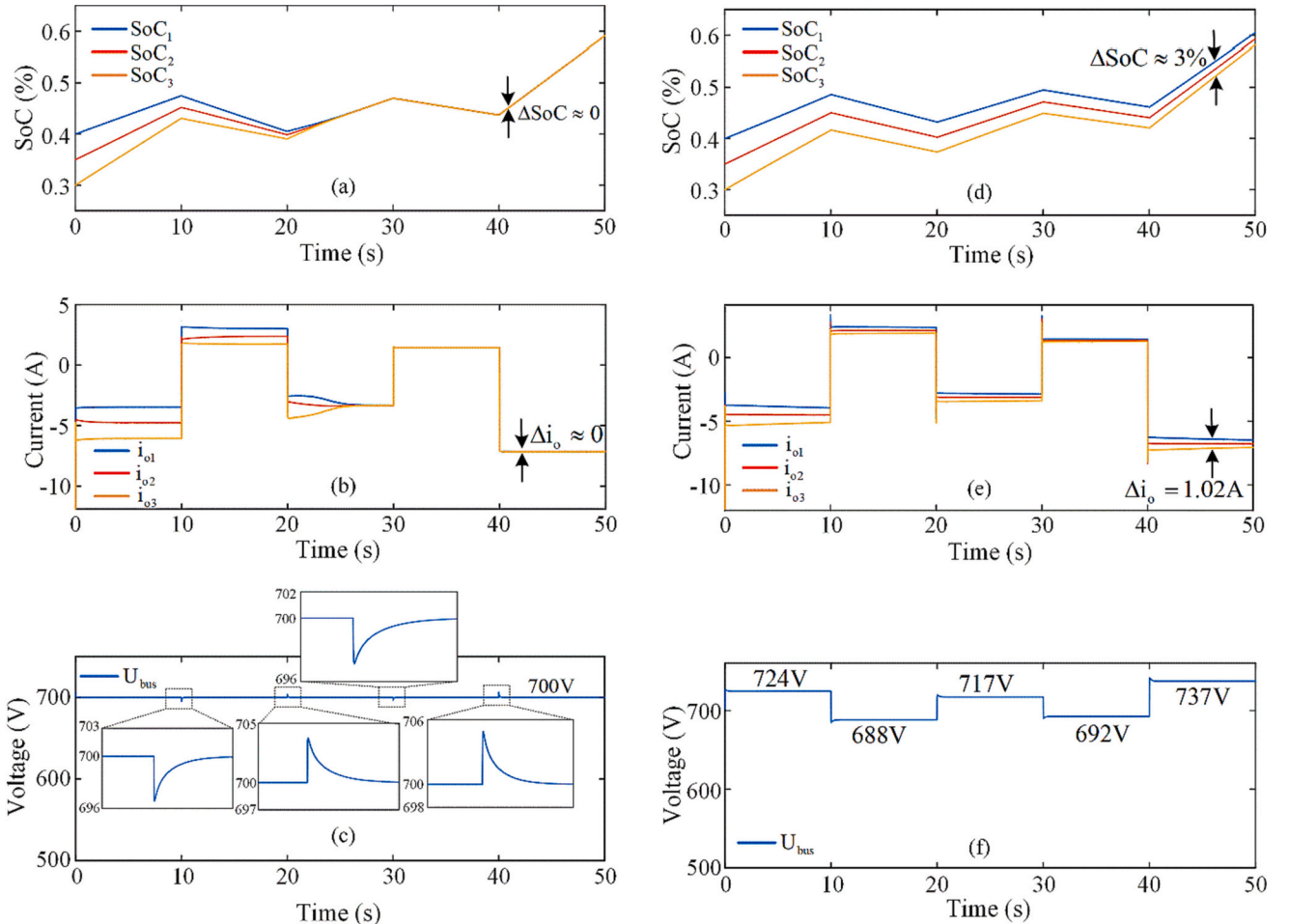


Fig. 13. Comparative simulation results (a), (b), (c) of proposed method and (d), (e), (f) of ref. method [32].

Phase II (10 to 20s): The PV array generates 25 kW of total power, while the total load power is 30 kW.

Phase III (20 to 30s): The PV power remains constant, consistent with phases I and II, while the load power decreases from 30 kW to 18 kW.

Phase IV (30 to 40s): In this phase, both PV power and load power undergo simultaneous changes, with PV power set at 12 kW and load power at 30 kW, respectively.

Phase V (40 to 50s): The PV array power and load power are 30 kW and 15 kW, respectively.

Fig. 13 illustrates the simulation outcomes for Case IV. Fig. 13(a), (b), and (c) display the simulation results of the proposed methodology, while Fig. 13(d), (e), and (f) depict the simulation results of the reference approach in [32]. When implementing the control strategy described in Ref. [32], the mismatch in line impedance fails to achieve the specified control objectives within the designated simulation time-frame. In conclusion, there are deviations of approximately 3 % SoC equilibrium and 1.02 A in current distribution. Additionally, the bus voltage experiences a drop of approximately 24 V due to the absence of voltage drop compensation measures. In contrast, it is revealed that the method presented in this paper achieves SoC equilibrium and accurate load current distribution at approximately $t = 24$ s, even when factors such as line impedance mismatch and voltage drop are considered. Additionally, when we compare Fig. 13(c) and (e), it is apparent that employing the technique proposed in this study results in the ability to stabilize the bus voltage to its predefined target, even in situations where there are sudden fluctuations in load power or PV power. In contrast, when the approach of Ref. [32] is employed, the bus voltage deviates from the given value by approximately 24 V throughout the entire simulation process. The simulation results for this case study demonstrate the superior performance of the proposed control scheme compared to the method described in Ref. [32]. In this context, the SoCs rapidly converge to a consistent state, accurate load current sharing is achieved, even in the presence of line impedance variations, and effective compensation is provided for bus voltage.

The above comparative analysis confirms that the proposed method demonstrates significantly superior performance. The SoC balancing

and current sharing of each BSU have ultimately attained consistency, with errors approximately around 0 % and 0 A. In contrast, the comparison method results in the SoC equilibrium and current sharing of each BSU with errors approximately around 3 % and 1.02 A.

5.5. Case study V: evaluation of the proposed strategy with communication delay

In this scenario, the primary aim is to evaluate the effectiveness of the proposed strategy in cases involving communication delays, considering conditions of photovoltaic (PV) and load switching. The initial SoC for each ESU is set at 50 %, 45 %, and 40 %, respectively, and the ratio of the rated capacity for each ESU remains uniform with C1:C2:C3 = 2:2:2. The three-stage simulation process remains consistent with scenario III. Fig. 14 presents the simulation process for case study V. Fig. 14(a), (b), and (c) present the simulation results without communication delay, while Fig. 14(d), (e), and (f) depict the simulation results with an 80 ms communication delay. As observed, the achievement of the corresponding control objectives remains intact even in the presence of communication delays. If the communication delay is 80 ms, a transient oscillation time of 350 ms is observed in Fig. 14(d), (e), and (f), while the system without communication delay remains stable without oscillations. Simultaneously, the voltage drops for both scenarios can be regulated within 5 V and 2 V, respectively. Nevertheless, even with the presence of communication delays, our proposed scheme achieves SoC equilibrium and precise load current distribution around $t = 20$ s. By this time, the SoC and current of each ESU have ultimately attained consistency, with errors approximately around 0 A and 0 %, respectively.

5.6. Case study VI: evaluation of proposed strategy under low and high SoC level condition

This scenario primarily assesses the control performance of the proposed scheme approach under low and high SoC level condition during the normal operation of the system. In the high level SoC charging phase, the initial SoC values for the three BSUs are set at 50 %, 55 %, and 65 %, respectively, while the PV system consistently generates

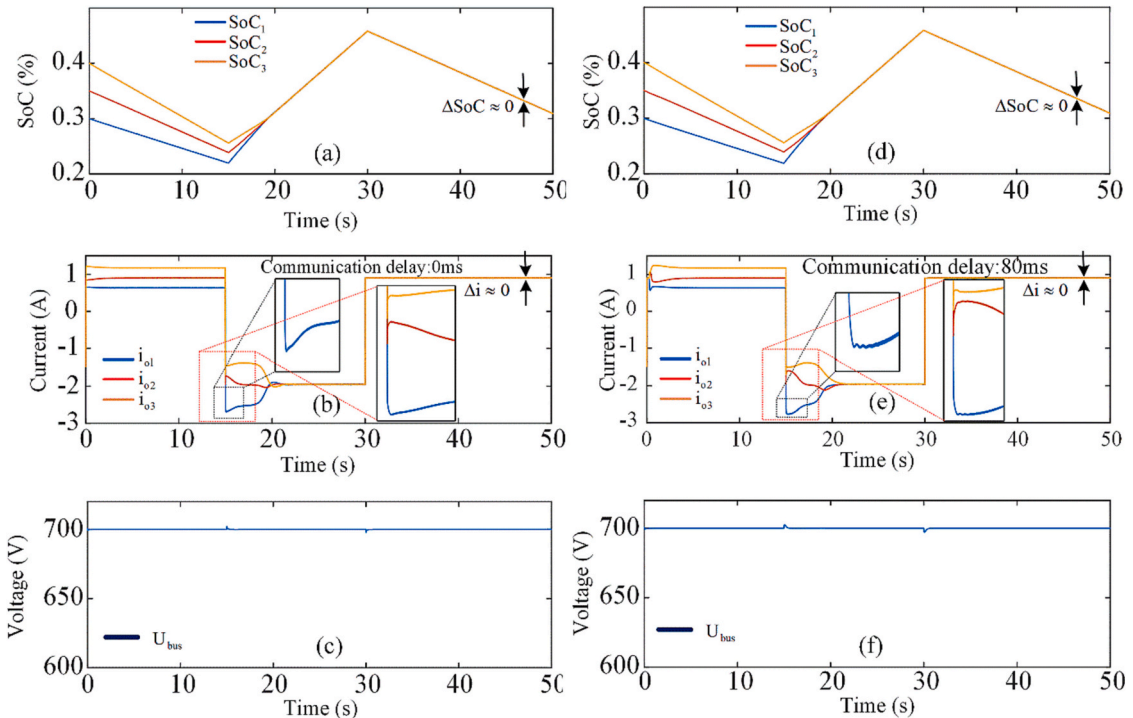


Fig. 14. Simulation results (a), (b), (c) of proposed method without communication delay and (d), (e), (f) of proposed method with communication delay.

5 kW of output power, and the controllable load power is maintained at 1 kW. During the low level SoC discharging phase, the initial SoC values for each BSU are set at 25 %, 35 %, and 45 %, respectively. The PV system output power is configured at 1 kW, with the load power remaining at 5 kW. Moreover, the rated capacity is consistently set at 2 Ah for all three BSUs throughout both phases. Fig. 15 depicts simulation results for the processes involving low and high SoC levels of the BSUs, where Fig. 15(a), (b), and (c) present results for the high-level SoC charging phase, and Fig. 15(d), (e), and (f) illustrate results for the low-level SoC discharge phase.

During the charging process, when the master BSU reaches its SoC upper limit, the remaining two BSUs need to undertake more charging power, and their SoC balancing is accelerated. When the second master BSU also reaches its SoC upper limit, the charging power of the remaining BSU increases further until it reaches its SoC upper limit and exits operation. However, when all BSUs reach their maximum SoC limit, they go out of operation. At this point, the PV system switches from Maximum Power Point Tracking (MPPT) mode to constant voltage mode to ensure the stability of the DC bus. During the discharging phase, when the SoC of BSU_3 reaches its lower limit, the remaining two BSUs must increase their discharging power to maintain system power balance, thereby accelerating SoC balancing. As the SoC of BSU_2 also reaches its lower limit, the discharging power of the remaining BSU_1 increases until it reaches its SoC lower limit and exits operation. However, when all BSUs reach their minimum SoC limit, they cease operation. At this point, the Photovoltaic system switches from MPPT mode to constant voltage mode, ensuring the stability of the DC bus. The results depicted in Fig. 15(c) and (d) illustrate smaller fluctuations and deviations in DC bus voltage. At the same time, the control mechanism ensures that the voltage drop is maintained within an acceptable range, usually below 3 V in both phases. This regulation is effective during the timeframe from 12 s–14 s in high-level SoC conditions and from 14 s–16 s in low-level SoC conditions.

The simulation outcomes validate that the proposed approach can maintain stable system operation even after the withdrawal of some BSUs, demonstrating its potential value for engineering applications.

6. Experimental results

To thoroughly test our proposed control strategy, we have developed

an experimental platform employing Star Sim HIL, as shown in Fig. (see Appendix A). This platform's core uses a HIL simulator, which is connected to the NI PXIe-FPGA-7846R. It includes important parts like three BSUs, three buck boost DC/DC converters, a solar panel array (PV), and a controllable load (CRL). Our experiments have been meticulously structured into four distinct scenarios. These scenarios meticulously assess the validity and effectiveness of the proposed approach under varying conditions, including fluctuating load demands, PV array output fluctuations, and challenges such as communication failures, and different rated capacity of BSU.

6.1. Case study I: evaluation of the proposed strategy during normal charge and discharge processes

In this experimental scenario, we evaluate the control performance of the proposed strategy under stable charging and discharging conditions for the BSUs within the normal system operation. During the charging phase, the initial SoC values for the three BSUs are configured at 25 %, 20 %, and 15 %, respectively. The PV system consistently generates 30 kW, and the controllable load power is maintained at 10 kW. As for the discharging phase, the initial SoC values for each BSU are set at 80 %, 75 %, and 70 %, correspondingly. The PV system output power is held steady at 5 kW, while the load power remains at 20 kW. Throughout both the charging and discharging phases, the rated capacity remains a constant 50 Ah for all three BSUs.

Fig. 16 presents experimental results for stable charging and discharging processes of the BSUs, with (a), (b), and (c) showing charging phase results and (d), (e), and (f) illustrating discharge phase results. In Fig. 16(a), (b), (d), and (e), it becomes evident that the implementation of the approach proposed in this paper results in the stable convergence of the SoC for each BSU, along with a gradual alignment of the output currents among the converters. At approximately $t = 90$ s in case of charging and around $t = 110$ s in case of discharging, we observe the achievement of SoC balance and synchronized current sharing among all converters, demonstrating a relatively high level of speed and precision in the current-sharing process. Moreover, as seen in Fig. 16(c) and (d), during the regular charging and discharging processes, the method presented in this paper demonstrates the ability to maintain the bus voltage at approximately 700 V.

These experimental results affirm the capability of the proposed

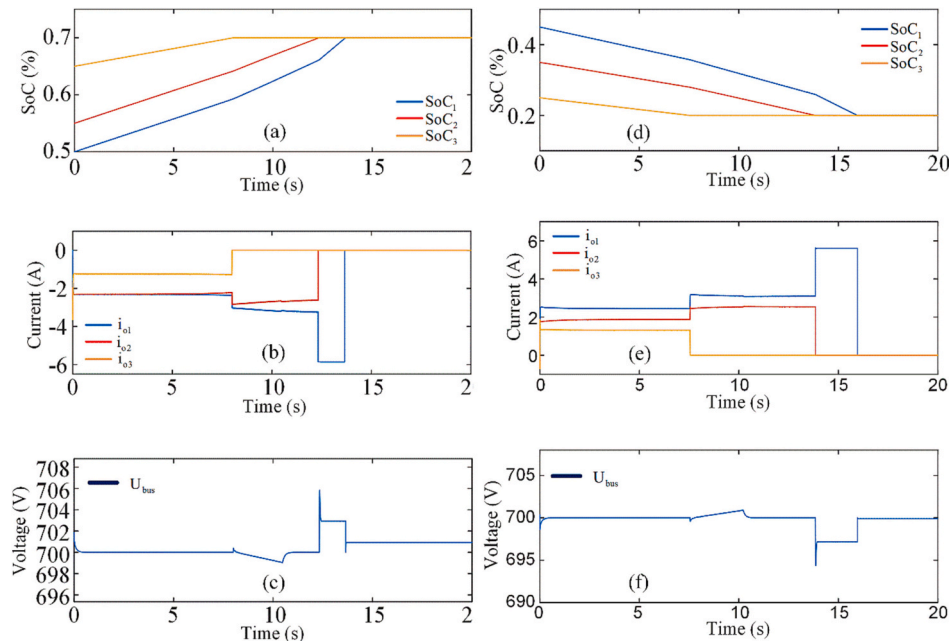


Fig. 15. Simulation result of low and high SoC level condition.

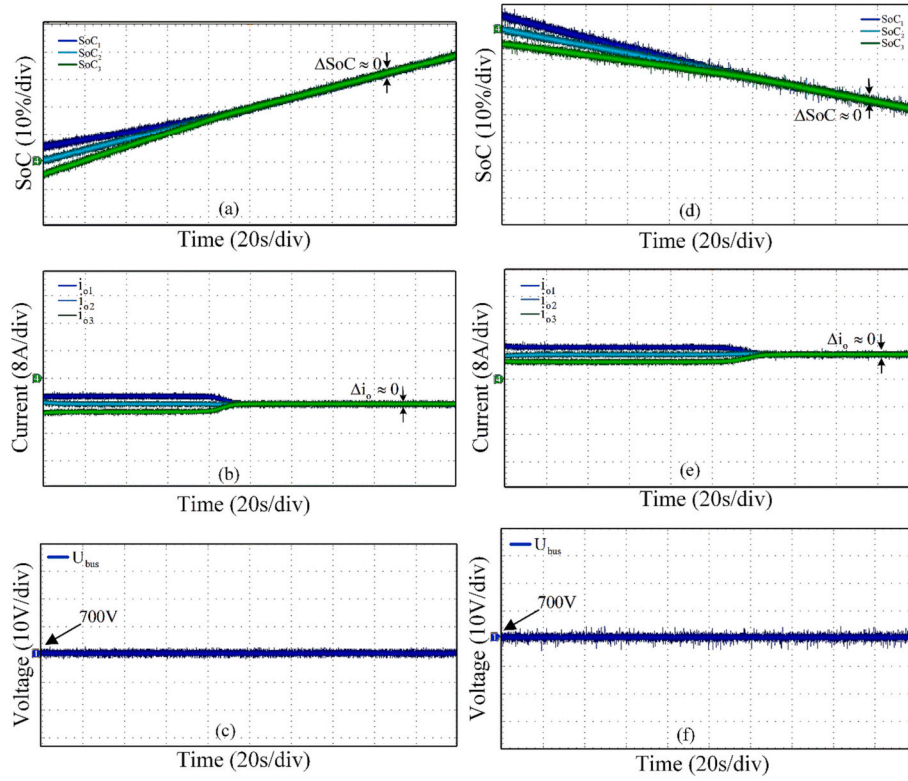


Fig. 16. Experimental result (a), (b), (c) of stable charging and (d), (e), (f) of discharging.

approach to attain SoC convergence swiftly and reliably during both charging and discharging phases of the BSS. Additionally, it ensures accurate load current sharing while keeping the bus voltage within acceptable bounds.

6.2. Case study II: evaluation of the proposed strategy under simultaneous sudden load and PV power changes

In evaluating the proposed strategy's performance, we focus on its response to sudden fluctuations in PV system output power and load power under normal communication conditions. The three BSUs, initially with SoC values of 50 %, 45 %, and 40 % and a 25 Ah rated capacity each, accommodate varying aging levels. The experimental

process comprises five phases:

Phase I (0–40s): PV power output is 40 kW, and load power is 15 kW.

Phase II (40–80s): PV power reduced to 10 kW, while load power abruptly increases to 30 kW.

Phase III (80–120s): PV power is increased to 30 kW, and load power decreases from 30 kW to 15 kW.

Phase IV (120–160s): Both PV power (12 kW) and load power (30 kW) change simultaneously.

Phase V (160–200s): PV power reaches 42 kW, and load power drops to 15 kW.

Fig. 17 presents an overview of the experimental results in response to abrupt fluctuations in both PV system output power and load power.

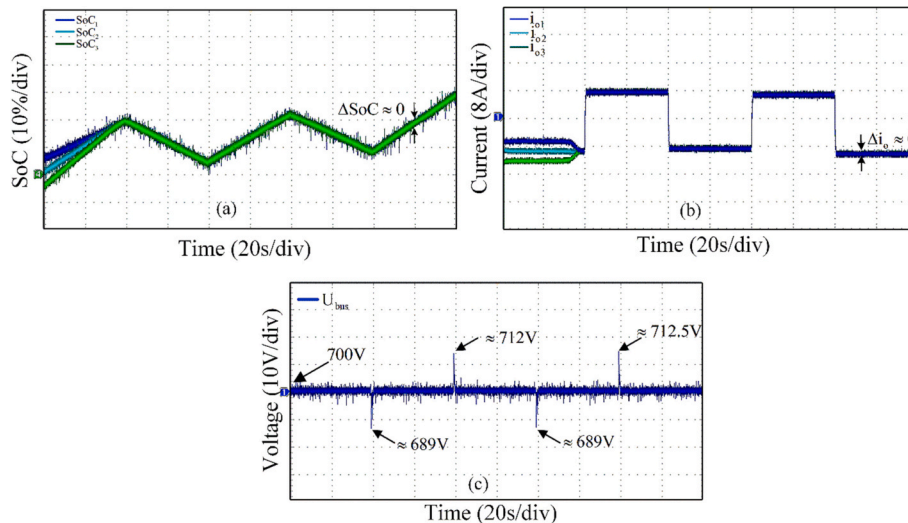


Fig. 17. Experimental result of PV fluctuation and sudden load changes.

At $t = 0$ to 40s, the PV system generates surplus power compared to the load demand, prompting each BSU to switch to charging mode to capture the excess energy. However, at $t = 40$ s, the PV system's output falls short of the load demand, leading the BSUs to transition from charging mode to discharging mode to compensate for the deficit. At $t = 80$ s, a sudden reduction in load power to 15 kW prompts the BSUs to revert to charging mode to maintain power balance, resulting in a gradual convergence of SoC and load current during this interval. At $t = 120$ s, the PV system's output unexpectedly drops to 12 kW, while the load increases to 30 kW. In response, each BSU switches to discharging mode to mitigate the power deficit. At $t = 160$ s, the PV system's output surges to 42 kW, accompanied by a decrease in load power to 15 kW, prompting each BSU to return to charging mode to store excess power.

Throughout the entire experimental timeframe, our approach consistently achieved SoC equilibrium and accurate load current distribution, notably reaching convergence around $t = 37$ s. In scenarios involving rapid changes in load power, PV power, or both, we observed fluctuations in bus voltage. However, our proposed method effectively restored the bus voltage to 700 V, demonstrating improved dynamic performance in both voltage and current during transient power fluctuations.

The result of above analysis confirms that the proposed scheme achieves faster equalization speed and higher accuracy. The dynamic effect of charge and discharge switching for each BSU is excellent, and it can maintain the bus voltage within acceptable limits.

6.3. Case study I: evaluation of the proposed strategy with communication failure and different capacity

In our experimental investigation, we assessed the proposed strategy's performance with different rated capacities of BSUs and communication failures. Each BSU started with initial SoC levels of 40 %, 35 %, and 30 %, corresponding to rated capacities of 10 Ah, 20 Ah, and 30 Ah. We divided the experiment into five phases:

Phase I (0–40s): PV output power remained at 35 kW, with a 10 kW load. Communication between BSU_1 and BSU_2 failed from 0 to 6 s.

Phase II (40–80s): PV output reduce to 5 kW, and the load increased to 20 kW.

Phase III (80–120s): PV output increased from 5 kW to 30 kW, while the load dropped to 10 kW.

Phase IV (120–160s): Both PV power and load are set 12 kW and 20 kW, respectively.

Phase V (160–200s): PV power reached 36 kW, while the load decreased from 20 kW to 10 kW.

Our experimental results, displayed in Fig. 18(a), (b), and (c),

demonstrate the strategy's robustness, even in the presence of communication failures. This resilience is attributed to the continuous existence of a minimum spanning tree within the communication system, facilitating consistent variable estimation by the MACA algorithm. At approximately $t = 27$ s, the system achieves SoC equilibrium, maintains proportional load current distribution, and sustains an output current ratio of 10:20:30, with minimal SoC balance error. Additionally, the bus voltage can swiftly restore itself to a stable state and consistently maintain its rated value in the face of disturbances. In summary, our experimental findings in scenario III validate the effectiveness of our approach in addressing sudden communication failures.

The above analysis confirms that the proposed approach can still maintain normal system operation and ensure the continuous convergence of the SoC for each BSU and proportional load current distribution, even in the presence of a communication failure, with different rated capacities for each BSU.

6.4. Case study IV: comparison with advance method

In our experimental evaluation, we compare the proposed methodology with the approach presented in Ref. [32], focusing on scenarios involving load and PV fluctuations. It's important to note that the methodology presented in Ref. [32] fails to account for crucial factors such as line impedance compensation, and bus voltage restoration, resulting in slower SoC equilibrium and less accurate load current distribution. To highlight our approach's advantages, we compare it with the method from Ref. [32]. Initial SoC levels for each BSU are 45 %, 40 %, and 35 %, with uniform 20 Ah rated capacity of each BSU. We also set parameter p to 4 for the SoC equalization method from Ref. [32]. The experiment comprises five phases:

Phase I (0–40s): Initial PV and load power are 35 kW and 15 kW, respectively, with both models activated at $t = 0$ s.

Phase II (40–80s): PV generates an 8 kW power, while load power rises to 30 kW.

Phase III (80–120s): PV power change to 30 kW as load power drops from 30 kW to 15 kW.

Phase IV (120–160s): Both PV and load power change simultaneously, with PV at 12 kW and load at 30 kW.

Phase V (160–200s): PV and load power are set at 38 kW and 15 kW, respectively.

Fig. 19 illustrates the results, with Fig. 19(a), (b), and (c) showing our methodology and Fig. 19(d), (e), and (f) displaying the Ref. [32] approach. The Ref. [32] strategy, due to line impedance mismatches, fails to meet control objectives, resulting in about 4 % SoC equilibrium deviation and 2 A current distribution variance. Bus voltage drops by

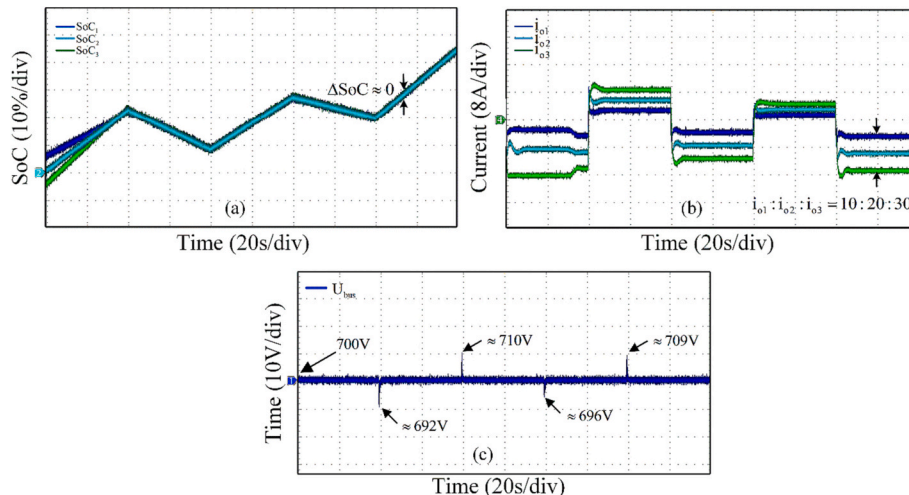


Fig. 18. Experimental results of communication failure and different capacity.

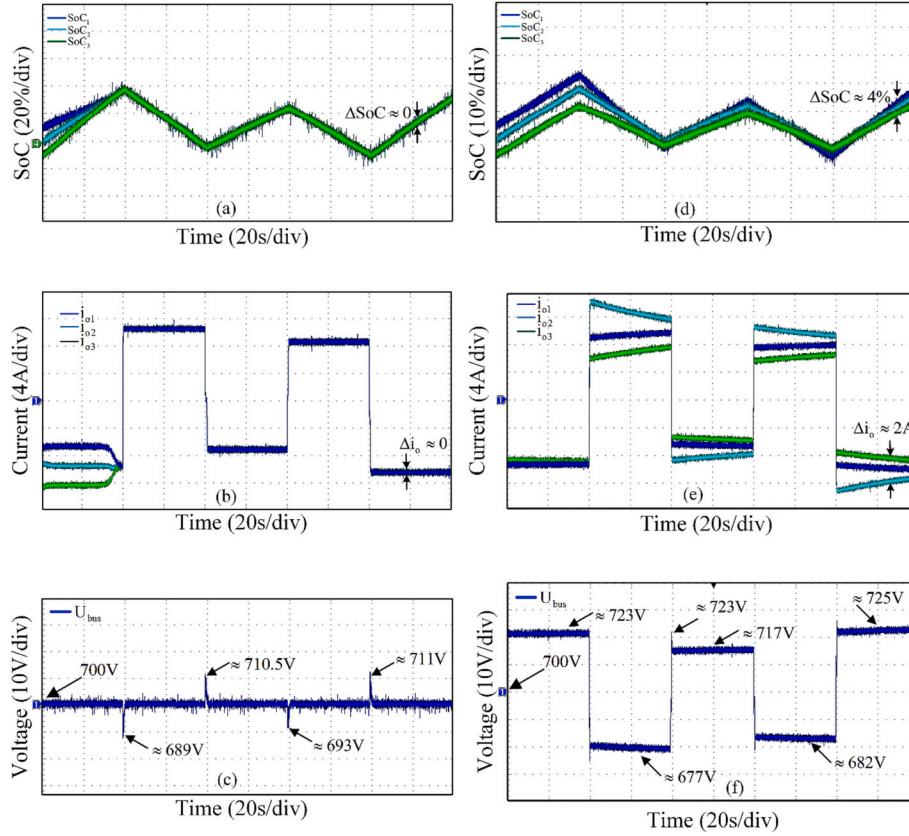


Fig. 19. Comparative experimental results (a), (b), (c) of proposed method and (d), (e), (f) of ref. method [32].

approximately 23 V due to the absence of voltage drop compensation, as seen in Fig. 19(e). In contrast, our method achieves SoC equilibrium and precise load current distribution around $t = 28$ s, even with line impedance mismatches and voltage fluctuations. Notably, our approach stabilizes bus voltage, as seen in Fig. 19(c), even during load or PV power fluctuations. In contrast, using Ref. [32] results in an average bus voltage deviation of approximately 23 V.

The above comparative analysis confirms this case study demonstrates the superiority of our control scheme compared to Ref. [32], with rapid SoC convergence, accurate load current sharing, and effective compensation for voltage fluctuations. The SoC balancing and current sharing of each BSU have ultimately attained consistency, with errors approximately around 0 % and 0 A. In contrast, the comparison method yields SoC equilibrium and current sharing for each BSU with errors approximately around 4 % and 2 A.

7. Conclusion

To address the issues of unbalanced SoC, inaccurate current distribution, and DC bus voltage deviations, this paper proposes an adaptive droop based SoC control approach for distributed BSSs in DC microgrids. Within this approach, the sigmoid function is integrated as the carrier, enabling adaptive adjustment of the droop coefficient based on the initial SoC value of the BSU. Furthermore, the characteristics of the sigmoid function constrain the droop coefficient within permissible limits. To mitigate the effects of line impedance and facilitate bus voltage recovery, virtual voltage drop equalization and bus voltage compensation controllers are introduced in this study. To reduce communication burdens, a multiagent consensus algorithm is employed in the communication layer. Furthermore, a stability analysis is conducted in this study to provide theoretical support for the proposed scheme. Finally, the effectiveness and reliability of the proposed scheme

have been verified through MATLAB/Simulink simulations and experimental based platform (Star Sim HIL). Several challenges persist within this field, such as the need to develop more streamlined distributed control approaches. In our future research, we will continue to enhance the simplicity of the control scheme's design. Moreover, we will undertake empirical validation of the proposed control scheme by applying it to an operational ship.

CRediT authorship contribution statement

Rashid Iqbal: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Yan-cheng Liu:** Project administration, Resources, Supervision, Validation. **Yuji Zeng:** Writing – review & editing. **Qinjin Zhang:** Writing – review & editing. **Heyang Yu:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

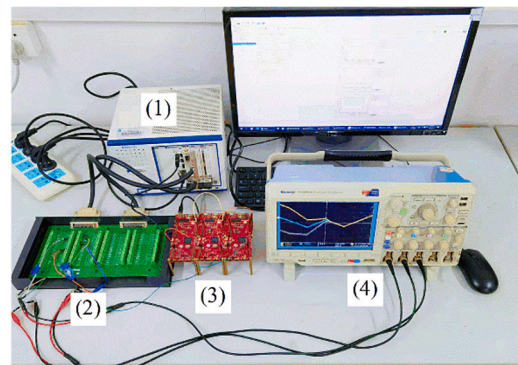
Data availability

No data was used for the research described in the article.

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Appendix A



(1) Hardware in the loop real time simulator
(2) IO Interface (3) Converter controllers (4) Oscilloscope

Fig. Experimental setup.

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